

Eastern International University

Becamex Business School



ASSIGNMENT 2

PORFOLIO SITE

Course: MIS 311 – Introduction to Business Analytics

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1. Data Overview

Table 1. Dataset Overview

Item	Description
Dataset Name	Shoe Price Dataset
Source	Dataset provided by the lecturer for MIS311 – Introduction to Business Analytics
Number of Rows	203
Number of Columns	4
Variables	Brand, Color, Size, Price
Context	The dataset contains information on different shoe products, including their brand, color, size, and selling price. It is designed for exploratory data analysis to identify pricing patterns and product characteristics.

Table 2. Variable Description

Variable	Data Type	Description
Brand	Object	Brand name of the shoe product
Color	Object	Color of the shoe product
Size	Float	Shoe size
Price	Float	Selling price of the shoe product

2. Data Cleaning

Before conducting the analysis, the dataset was cleaned by checking for missing values and duplicate records to improve data quality.

2.1. Missing Values

Table 3. Missing Values Summary

Variable	Missing values	Percentage	Action taken
Brand	0	0.00%	No action required
Color	1	0.49%	Removed the affected record
Size	0	0.00%	No action required
Price	1	0.49%	Removed the affected record

The dataset contains 203 records. One missing value was identified in the Price column (Row 23) and one missing value was identified in the Color column (Row 192). As each missing value represented only 0.5% of the dataset, the affected records were removed before conducting the analysis.

2.2. Duplicate Records

Table 4. Duplicate Records Summary

Item	Result
Duplicate Rows Found	3
Action Taken	Removed all duplicate records

The dataset was checked for duplicate records using the Remove Duplicates function in Microsoft Excel. A total of three duplicate rows were identified and removed. Removing duplicate records helps prevent repeated observations from affecting the descriptive statistics and ensures that the analysis is based on unique data entries.

3. Descriptive Statistics

Descriptive statistics were calculated to summarize the key characteristics of the shoe prices. These statistics provide an overall understanding of the data distribution and price variation within the dataset.

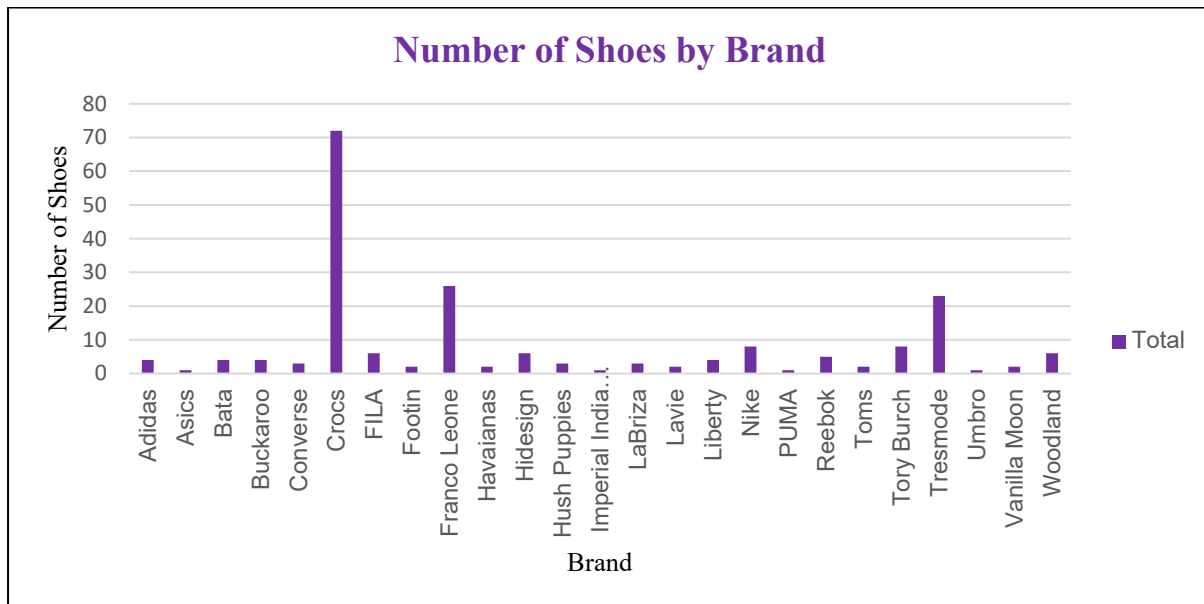
Table 5. Descriptive Statistics of Shoe Prices

Statistics	Value
Mean	212.57
Median	212
Mode	347
Standard Deviation	79.22
Minimum	80
Maximum	350
Range	270
Count	198

Table 5 presents the descriptive statistics of shoe prices after data cleaning. The average shoe price is 212.57, while the median price is 212, indicating that the average price is very close to the middle value of the dataset. Prices range from 80 to 350, with a standard deviation of 79.22, suggesting a moderate variation in shoe prices across different brands.

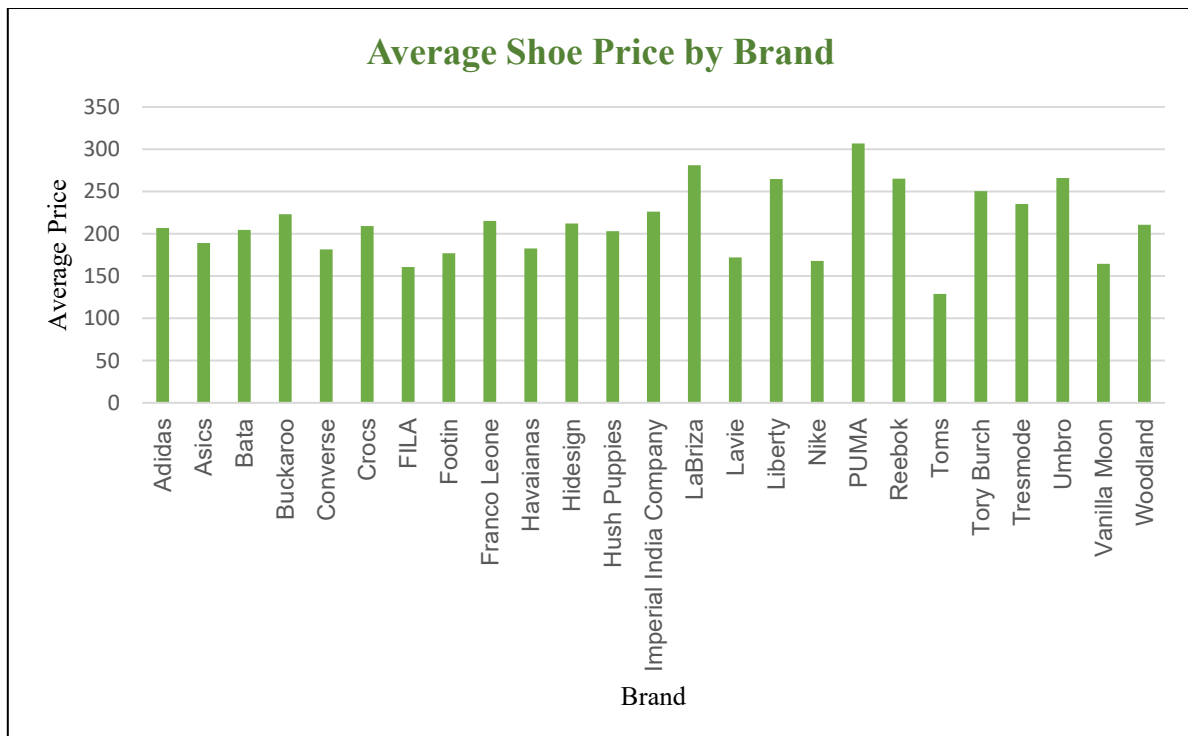
4. Visualizations

Figure 1. Number of Shoes by Brand



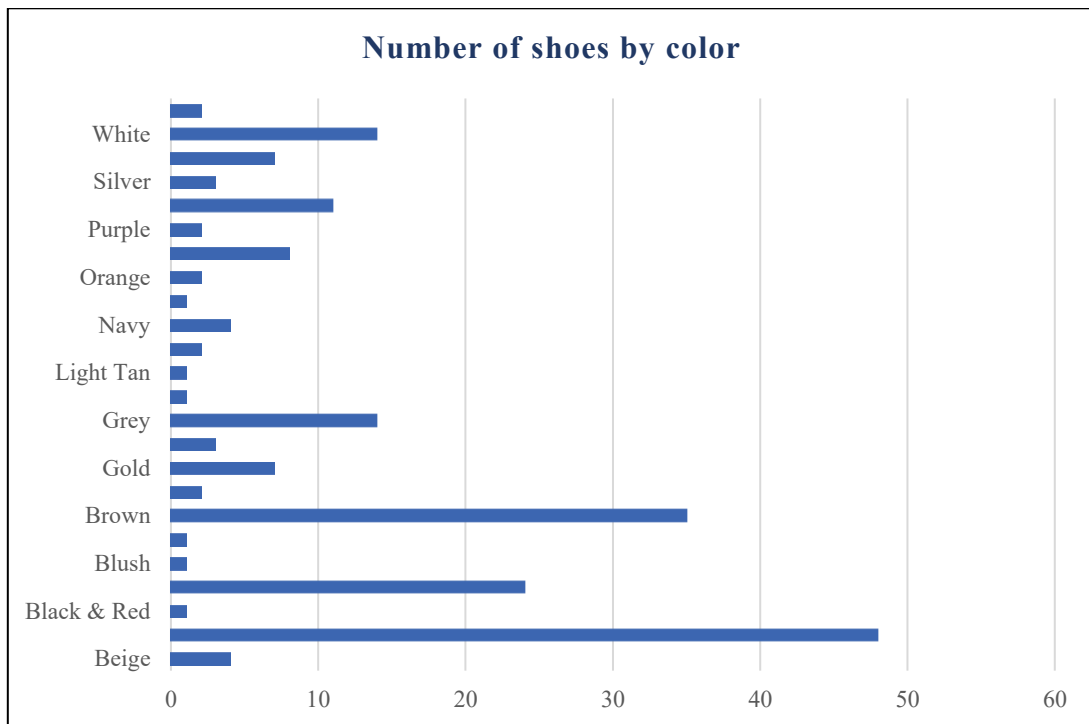
Crocs has the highest number of products in the dataset, followed by Franco Leone. Most other brands contain relatively few products, indicating an uneven distribution of products among brands.

Figure 2. Average Shoe Price by Brand



The average selling price varies noticeably across brands. For example, Reebok has one of the highest average prices, whereas Toms has one of the lowest. This suggests that brands adopt different pricing strategies to target distinct customer segments.

Figure 3. Number of Shoes by Color



This figure shows the distribution of shoe products by color. Black is the most common color, followed by Brown and Blue, indicating that neutral and classic colors dominate the product range. These colors are generally more versatile and may better match customer preferences.

5. Key Insights

Insight 1

The mean price (212.57) is almost identical to the median price (212.00), indicating that shoe prices are fairly balanced and are not heavily affected by extreme values. This suggests that the dataset represents a relatively stable pricing distribution.

Insight 2

Crocs has the largest number of products (72 products) in the dataset, while average prices differ noticeably across brands. This indicates that brands compete through both product variety and pricing strategies to attract different customer segments.

6. Conclusion

This report analyzed the Shoe Price dataset using descriptive statistics and data visualization techniques. After cleaning the data, several patterns in product distribution and pricing were identified. These findings demonstrate how exploratory data analysis can support business understanding and provide useful insights for decision-making.